INTRODUCTION TO OOP 2024-2025 PROJECT REPORT

Hasan Can Yıldız– 05230000394 Fatih Uzer – 05230001160

Player Management System using Doubly Linked List and Stack

Group Members : Hasan Can Yıldız– 05230000394
Fatih Uzer – 05230001160

INTRODUCTION TO OOP 2024-2025 PROJECT REPORT

Hasan Can Yıldız– 05230000394 Fatih Uzer – 05230001160

INTRODUCTION

In this project, we develop an information system for football players using a doubly linked list data structure. The linked list must be listed alphabetically by player name and surname. The application should print all players' information (name, surname, date, clubs) in ascending alphabetical order or in descending alphabetical order.

PROGRAMMER CATALOG

We use 5 classes for the project:

1)Date Class: For the date of birth,we used a variable named "**birthDate"**in the "Date" class type given in our textbook ("Absolute Java") .

2)Player Class: This class represents a football player with attributes and it uses toString method to display player's information in a string form.

3)DoublyLinkedList Class: It manages a list of football players using nodes that point both forward and backward.Also it can add a new player in alphabetical order (insert), deleting a player by name and surname (delete), and printing the list in both ascending and descending order (printAscending , printDescending).

4) Stack Class: This class implements an array-based stack that stores objects of type Object. It provides push (adding), pop (removing) and isEmpty (checking if the stack is empty). Copying the list into the stack and printing it in LIFO order is done in Main.

5)Main Class: This class reads the player data from players.txt and presents a menu with 8 options to the user.

Option 1. :Opens the players.txt file and reads all players' information. (loadFile)

Option 2. : Allows the user to add a new player manually. (addPlayer)

Option 3. : Displays the player's information based on the given name and surname. (showPlayer)

Option 4. : Deletes a player from the list by name and surname. (delete)

Option 5. : Displays the list in ascending order. (printAscending)

Option 6. : Displays list in descending order. (printDescending)

Option 7. : Copies the list into the stack and displays it. (push, pop, isEmpty)

Option 8. : Saves the list to a file and exits the program. (saveFile)

TIME SPENT

Design and planning: 1.5 hours

Implementation (classes and main logic): 10 hours

Testing and debugging: 2 hours

Report and formatting: 2 hours

Total: 15.5 hours